Create Capture History Objects

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Contents

Building Capture Histories for SCR Analyses **1**

Load data, source function 1

Closed population models 2

Open population models 6

Save the output 8

Building Capture Histories for SCR Analyses

In this markdown, we'll create capture histories for the SCR analyses that follow. Each object will include i) a `y` array of capture histories (with `dim = c(i, j, k, t)`, individuals, traps, occasions, years), ii) a `pc` array (of similar dimensions) with previous capture information (for models with a local behavioral effect), and iii) a `clone` vector that has identifiers for each individual i (from the `finalClone` column of our `sampleMetadata` file). We want to build several of these objects, given interest in fitting different (open vs. closed) models.

1. For closed population models - data for two years saved separately (3-d array), separate for the two sexes as well
   - Females in 2023
   - Females in 2024
   - Males in 2023
   - Males in 2024
2. For open population models - two years combined (4-d array), but still modeling the sexes separately
   - Females (2023 and 2024)
   - Males (2023 and 2024)

Load data, source function

Load *i*) the sample metadata with clone ID information and *ii*) the `.Rda` with locations and yearly operational status of hair snares. Source in functions for *i*) building capture history and previous capture objects and *ii*) calculating summaries of capture histories from a 3-d or 4-d `y` array (based on a function from the `scrbook` R package).

```r
#Load sample metadata (smd)
smd <- read.csv("../sampleinfo/sampleMetaData_cloneID.csv", header = TRUE)
#Load snare location information
load("../sampleinfo/snareLocations.Rda")
ls()
```

```
## [1] "locs" "oper" "smd"
```

```r
#Source the makeCH function
source("../input/makeCH.R")
#Source the sumstats functions (for 3d and 4d arrays)
source("../input/sumstats_4d.R")
```

Closed population models

Splitting the dataset up by year and sex…

```r
#Start with all hair samples from 2023 and 2024
sampsALL <- smd[which(!is.na(smd$finalClone) &
                        smd$is.SCR == TRUE &
                        smd$year %in% c(2023,2024) &
                        smd$FILT == FALSE &
                        !is.na(smd$finalSEX)),]

#Check the dataset
nrow(sampsALL)
```

```
## [1] 1509
```

```r
table(sampsALL$region)
```

```
##
##  CGA
## 1509
```

```r
table(sampsALL$finalSEX)
```

```
##
##   F   M
## 792 717
```

```r
sum(!sampsALL$finalSEX %in% c("M","F"))
```

```
## [1] 0
```

```r
length(unique(sampsALL$finalClone[!is.na(sampsALL$finalClone)]))
```

```
## [1] 188
```

```r
sum(is.na(sampsALL$finalClone))
```

```
## [1] 0
```

```r
#Separately for males, females, 2023, and 2024
snares23 <- locs[oper[,1] == 1,]
snares24 <- locs[oper[,2] == 1,]

#M23
Msamps23 <- sampsALL[which(sampsALL$year == 2023 &
                       sampsALL$finalSEX == "M"),]
dim(Msamps23)
```

```
## [1] 257  20
```

```r
table(Msamps23$finalSEX)
```

```
##
##   M
## 257
```

```r
table(Msamps23$year)
```

```
##
## 2023
##  257
```

```r
length(unique(Msamps23$finalClone[!is.na(Msamps23$finalClone)]))
```

```
## [1] 61
```

```r
M23 <- makeCH(samps=Msamps23,
              traps=snares23,
              yrs=2023)
dim(M23$y)
```

```
## [1]  61 130   7
```

```r
dim(M23$pc)
```

```
## [1]  61 130   7
```

```r
length(M23$clone)
```

```
## [1] 61
```

```r
#M24
Msamps24 <- sampsALL[which(sampsALL$year == 2024 &
                       sampsALL$finalSEX == "M"),]
dim(Msamps24)
```

```
## [1] 460  20
```

```r
length(unique(Msamps24$cloneID[!is.na(Msamps24$cloneID)]))
```

```
## [1] 0
```

```r
M24 <- makeCH(samps=Msamps24,
              traps=snares24,
              yrs=2024)
dim(M24$y)
```

```
## [1]  44 138   7
```

```r
dim(M24$pc)
```

```
## [1]  44 138   7
```

```r
length(M24$clone)
```

```
## [1] 44
```

```r
length(unique(c(M23$clone,M24$clone)))
```

```
## [1] 90
```

```r
#F23
Fsamps23 <- sampsALL[which(sampsALL$year == 2023 &
                       sampsALL$finalSEX == "F"),]
dim(Fsamps23)
```

```
## [1] 357  20
```

```r
length(unique(Fsamps23$finalClone[!is.na(Fsamps23$finalClone)]))
```

```
## [1] 51
```

```r
F23 <- makeCH(samps=Fsamps23,
              traps=snares23,
              yrs=2023)
dim(F23$y)
```

```
## [1]  51 130   7
```

```r
dim(F23$pc)
```

```
## [1]  51 130   7
```

```r
length(F23$clone)
```

```
## [1] 51
```

```r
#F24
Fsamps24 <- sampsALL[which(sampsALL$year == 2024 &
                       sampsALL$finalSEX == "F"),]
dim(Fsamps24)
```

```
## [1] 435  20
```

```r
length(unique(Fsamps24$finalClone[!is.na(Fsamps24$finalClone)]))
```

```
## [1] 72
```

```r
F24 <- makeCH(samps=Fsamps24,
              traps=snares24,
              yrs=2024)
dim(F24$y)
```

```
## [1]  72 138   7
```

```r
dim(F24$pc)
```

```
## [1]  72 138   7
```

```r
length(F24$clone)
```

```
## [1] 72
```

```r
length(unique(c(F23$clone, F24$clone)))
```

```
## [1] 98
```

```r
#And the summary stats
rowSums(M23$y)
```

```
##  [1] 3 2 5 7 8 1 1 1 3 1 6 1 3 2 1 5 1 1 2 4 1 4 1 1 1 2 2 1 7 2 2 1 1 2 1 1 1 3
## [39] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 1 1 4 2 2 1 2 2 1
```

```r
sumstats_3d(M23$y)
```

```
##        ss.totN     ss.totcaps     ss.avgcaps ss.indwsprecap    ss.avgtraps
##      61.000000     123.000000       2.016393      22.000000       1.639344
```

```r
rowSums(M24$y)
```

```
##  [1]  1  1 26  1 25  5  5  9  4  6  5  4  1 13  1  1  3  1  5  1  1  1  1  1  1
## [26]  1  4  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
```

```r
sumstats_3d(M24$y)
```

```
##        ss.totN     ss.totcaps     ss.avgcaps ss.indwsprecap    ss.avgtraps
##      44.000000     145.000000       3.295455      13.000000       2.295455
```

```r
rowSums(F23$y)
```

```
##  [1]  6  2  6 15 10  3  6  6  5  3  1  6  3  2  1  2  1  3  1  1  1  1  1  1  1
## [26]  1  1  1  1  1  3  1  1  2  1  1  1  1  1  2  1  2  1  1  1  1  1  1  1  1
## [51]  1
```

```r
sumstats_3d(F23$y)
```

```
##        ss.totN     ss.totcaps     ss.avgcaps ss.indwsprecap    ss.avgtraps
##      51.000000     119.000000       2.333333      13.000000       1.450980
```

```r
rowSums(F24$y)
```

```
##  [1]  9  5  6  7 10  2  6  2  1  5  3  1  2  5  2  2  1  2  4  2  1  1  1  1  1
## [26]  6  1  1  3  1  1  1  1  1  1  2  1  2  1  4  1  1  3  1  1  2  1  2  1  1
## [51]  1  1  2  1  1  2  1  2  1  1  1  4  1  2  1  1  2  1  1  1  1  1
```

```r
sumstats_3d(F24$y)
```

```
##        ss.totN     ss.totcaps     ss.avgcaps ss.indwsprecap    ss.avgtraps
##      72.000000     153.000000       2.125000      19.000000       1.388889
```

Open population models

Combining data for years, but sexes are still separate…

```r
#Females - both years
Fsamps <- sampsALL[which(sampsALL$finalSEX == "F"),]
length(unique(Fsamps$finalClone[!is.na(Fsamps$finalClone)]))
```

```
## [1] 98
```

```r
Fch <- makeCH(samps=Fsamps,
              traps=locs,
              yrs=c(2023,2024))
dim(Fch$y)
```

```
## [1]  98 154   7   2
```

```r
dim(Fch$pc)
```

```
## [1]  98 154   7   2
```

```r
length(Fch$clone)
```

```
## [1] 98
```

```r
#Males - both years
Msamps <- sampsALL[which(sampsALL$finalSEX == "M"),]
length(unique(Msamps$finalClone[!is.na(Msamps$finalClone)]))
```

```
## [1] 90
```

```r
Mch <- makeCH(samps=Msamps,
              traps=locs,
              yrs=c(2023,2024))

dim(Mch$y)
```

```
## [1]  90 154   7   2
```

```r
dim(Mch$pc)
```

```
## [1]  90 154   7   2
```

```r
length(Mch$clone)
```

```
## [1] 90
```

```r
#And the summary stats
rowSums(Fch$y)
```

```
##  [1] 13  8 15 25 10  5  3  7  6  7  7  5  3  1  6  1  5  5  4  1  6  2  5  3  2
## [26]  4  3  3  2  2  1  1  1  1  1  1  6  2  1  5  3  5  1  1  1  1  1  1  2  1
## [51]  1  1  2  1  1  1  1  3  1  1  1  1  1  3  1  1  2  1  1  1  2  2  2  1  1
## [76]  1  2  1  1  2  1  1  2  1  1  1  5  1  2  1  1  2  1  1  1  1  1  1
```

```r
sumstats_4d(Fch$y)
```

```
##        ss.totN     ss.totcaps     ss.avgcaps ss.indwsprecap    ss.avgtraps
##      98.000000     272.000000       2.775510      30.000000       1.540816
##      ss.avgyrs
##       1.255102
```

```r
rowSums(Mch$y)
```

```
##  [1]  1  8  2 14 13 33  1  1  1  1  3  1 11  1 29  1  2  1  6  1  1  2  7  7  1
## [26]  4  1  5  4  1  3  2  1  7  2  5 15  1  1  1  1  1  2  5  1  1  1  1  3  1
## [51]  1  1  1  1  1  1  1  1  1  1  1  1  4  1  1  1  1  1  1  1  1  1  1  1  2
## [76]  1  1  1  1  2  1  2  3  1  1  1  1  1  1  2
```

```r
sumstats_4d(Mch$y)
```

```
##        ss.totN     ss.totcaps     ss.avgcaps ss.indwsprecap    ss.avgtraps
##      90.000000     268.000000       2.977778      27.000000       2.077778
##      ss.avgyrs
##       1.166667
```

Save the output

Save all of these objects (`F23`,`F24`,`M23`,`M24`,`Fch`,`Mch`) to a single `.Rda` file…

```r
save(F23, F24, M23, M24, Fch, Mch, file = "CGA_CH.Rda")
```